

ECON 695 – Problem Set 4

Due: October 27, 2025

Question 1: Average Treatment Effects

Suppose we have a random sample of size N of which $N_1 < N$ were randomly assigned to the treatment group $D_i = 1$, and for these individuals we observe $Y_i = Y_i(1)$, while the rest $N_2 = N - N_1$ were assigned to the control group $D_i = 0$, for whom we observe $Y_i = Y_i(0)$.

1. Show that under random assignment of treatment, the ATE ($\alpha = E[Y_i(1) - Y_i(0)]$) is equal to another causal effect of interest, which is Average Treatment on the Treated ($ATT := E[Y_i(1) - Y_i(0) \mid D_i = 1]$). Generally speaking, how would you interpret ATE versus ATT?

Solution (Part 1). **Goal:** Show that, under random assignment of treatment,

$$\text{ATE} := E[Y_i(1) - Y_i(0)] \quad \text{equals} \quad \text{ATT} := E[Y_i(1) - Y_i(0) \mid D_i = 1].$$

Assumptions.

- (A1) *Random assignment (unconfoundedness):* $(Y_i(1), Y_i(0)) \perp D_i$.
- (A2) *Overlap/positivity:* $0 < \Pr(D_i = 1) < 1$.
- (A3) *SUTVA/consistency:* Each unit has well-defined potential outcomes and the observed outcome satisfies $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$.

Key implication of (A1). For any integrable function g ,

$$E[g(Y_i(1), Y_i(0)) \mid D_i = d] = E[g(Y_i(1), Y_i(0))], \quad d \in \{0, 1\}.$$

In particular,

$$E[Y_i(1) \mid D_i = 1] = E[Y_i(1)], \quad E[Y_i(0) \mid D_i = 1] = E[Y_i(0)].$$

Derivation. Using linearity of expectation,

$$ATT = E[Y_i(1) - Y_i(0) \mid D_i = 1] = E[Y_i(1) \mid D_i = 1] - E[Y_i(0) \mid D_i = 1].$$

By random assignment (A1),

$$E[Y_i(1) \mid D_i = 1] = E[Y_i(1)], \quad E[Y_i(0) \mid D_i = 1] = E[Y_i(0)].$$

Therefore,

$$ATT = E[Y_i(1)] - E[Y_i(0)] = E[Y_i(1) - Y_i(0)] = ATE.$$

Interpretation (ATE vs. ATT).

- $ATE = E[Y_i(1) - Y_i(0)]$ is the average causal effect in the full population.
- $ATT = E[Y_i(1) - Y_i(0) \mid D_i = 1]$ is the average causal effect among those assigned to treatment.

Under random assignment, treated and untreated groups are (in distribution) the same ex ante, so the average effect among the treated equals the population average effect: $ATT = ATE$. Without random assignment, these generally differ because D_i may be related to $(Y_i(1), Y_i(0))$ (selection).

2. Show:

$$E[Y_i D_i] = E[Y_i(1) \mid D_i = 1] \times \Pr(D_i = 1)$$

$$E[Y_i(1 - D_i)] = E[Y_i(0) \mid D_i = 0] \times \Pr(D_i = 0)$$

Solution (Part 2). We use two ingredients throughout: (i) the Law of Iterated Expectations (LIE), $E[W] = E[E[W \mid D_i]]$, and (ii) consistency/SUTVA, $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$.

First identity: $E[Y_i D_i] = E[Y_i(1) \mid D_i = 1] \times \Pr(D_i = 1)$.

$$\begin{aligned} E[Y_i D_i] &= E[E[Y_i D_i \mid D_i]] && \text{(LIE)} \\ &= E[D_i E[Y_i \mid D_i]] && \text{(pull out } D_i, \text{ measurable w.r.t. } D_i) \\ &= \Pr(D_i = 1) E[Y_i \mid D_i = 1] && \text{(only the } D_i = 1 \text{ term survives)} \\ &= \Pr(D_i = 1) E[Y_i(1) \mid D_i = 1] && \text{(consistency when } D_i = 1). \end{aligned}$$

Second identity: $E[Y_i(1 - D_i)] = E[Y_i(0) \mid D_i = 0] \times \Pr(D_i = 0)$.

$$\begin{aligned}
E[Y_i(1 - D_i)] &= E[E[Y_i(1 - D_i) \mid D_i]] && \text{(LIE)} \\
&= E[(1 - D_i) E[Y_i \mid D_i]] && \text{(pull out } 1 - D_i) \\
&= \Pr(D_i = 0) E[Y_i \mid D_i = 0] && \text{(only the } D_i = 0 \text{ term survives)} \\
&= \Pr(D_i = 0) E[Y_i(0) \mid D_i = 0] && \text{(consistency when } D_i = 0).
\end{aligned}$$

These equalities require only LIE and consistency/SUTVA; random assignment is not needed for part (2).

3. Show that under random assignment of treatment,

$$\hat{\alpha} = \frac{\sum_{i=1}^N D_i Y_i}{\sum_{i=1}^N D_i} - \frac{\sum_{i=1}^N (1 - D_i) Y_i}{\sum_{i=1}^N (1 - D_i)}$$

is a consistent estimator for $\alpha = E[Y_i(1)] - E[Y_i(0)]$.

Solution (Part 3). Write the difference-in-means estimator as

$$\hat{\alpha} = \underbrace{\frac{\sum_{i=1}^N D_i Y_i}{\sum_{i=1}^N D_i}}_{\bar{Y}_1} - \underbrace{\frac{\sum_{i=1}^N (1 - D_i) Y_i}{\sum_{i=1}^N (1 - D_i)}}_{\bar{Y}_0} = \bar{Y}_1 - \bar{Y}_0,$$

where $\bar{Y}_1$ and $\bar{Y}_0$ are the sample means in the treated and control groups, respectively.

Assumptions. (i) Random assignment: $(Y_i(1), Y_i(0)) \perp D_i$; (ii) overlap: $0 < \Pr(D_i = 1) < 1$; (iii) SUTVA/consistency: $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$; (iv) finite first moments.

Key identities from consistency. $D_i Y_i = D_i Y_i(1)$, $(1 - D_i) Y_i = (1 - D_i) Y_i(0)$.

Limits of numerators and denominators (LLN). Let $p := \Pr(D_i = 1)$. By the Weak Law of Large Numbers (WLLN),

$$\frac{1}{N} \sum_{i=1}^N D_i \xrightarrow{p} p, \quad \frac{1}{N} \sum_{i=1}^N (1 - D_i) \xrightarrow{p} 1 - p.$$

Moreover, using the identities above and random assignment,

$$\frac{1}{N} \sum_{i=1}^N D_i Y_i = \frac{1}{N} \sum_{i=1}^N D_i Y_i(1) \xrightarrow{p} E[D_i Y_i(1)] = E[D_i] E[Y_i(1)] = p E[Y_i(1)],$$

$$\frac{1}{N} \sum_{i=1}^N (1 - D_i) Y_i = \frac{1}{N} \sum_{i=1}^N (1 - D_i) Y_i(0) \xrightarrow{p} E[(1 - D_i) Y_i(0)] = (1 - p) E[Y_i(0)].$$

The equalities $E[D_i Y_i(1)] = E[D_i] E[Y_i(1)]$ and $E[(1 - D_i) Y_i(0)] = E[1 - D_i] E[Y_i(0)]$ follow from $(Y_i(1), Y_i(0)) \perp D_i$.

Ratios and Slutsky/Continuous Mapping. Hence, using the sample means:

$$\begin{aligned} \bar{Y}_1 &= \frac{\frac{1}{N} \sum_{i=1}^N D_i Y_i}{\frac{1}{N} \sum_{i=1}^N D_i} \xrightarrow{p} \frac{p E[Y_i(1)]}{p} = E[Y_i(1)], \\ \bar{Y}_0 &= \frac{\frac{1}{N} \sum_{i=1}^N (1 - D_i) Y_i}{\frac{1}{N} \sum_{i=1}^N (1 - D_i)} \xrightarrow{p} \frac{(1 - p) E[Y_i(0)]}{1 - p} = E[Y_i(0)]. \end{aligned}$$

Therefore, by Slutsky's theorem,

$$\hat{\alpha} = \bar{Y}_1 - \bar{Y}_0 \xrightarrow{p} E[Y_i(1)] - E[Y_i(0)] = \alpha.$$

Thus, $\hat{\alpha}$ is a consistent estimator for the ATE under random assignment and overlap.

Question 2: Self Sufficiency Program (see ssp.ipynb)

Background from Card and Hyslop (2005):

- In the Self Sufficiency Project (SSP), members of a randomly assigned treatment group could receive a subsidy for full-time work. The subsidy was available for 3 years, but only to people who began working full time within 12 months of random assignment.
- SSP provides two incentives: (1) find a job and leave welfare within a year, and (2) choose work over welfare in the longer term.
- This question is designed to apply the IV-LATE framework to estimate the causal effects of SSP on leaving welfare, and characterize the compliers who would not have found a full-time job in the absence of the SSP subsidy. The data set `welfare.csv` contains 5,480 observations for people in the SSP experiment. The data set has the following variables:
 - `treatment` = 1 if in treatment group
 - `ft15`, `ft20`, `ft24`, `ft48` = set of dummies if working full time at months 15, 20, 24, and 48 after random assignment
 - `welfare15...welfare48` = set of dummies if receiving welfare at months 15, 20, 24, and 48 after random assignment
 - Other characteristics:
 - `imm` = 1 if immigrant
 - `hsgrad` = 1 if high school graduate (or higher) education
 - `age1t25` = 1 if age is 25 or less
 - `age35p` = 1 if age is 35 or more
 - `working_at_baseline` = 1 if person was working when assigned to treatment/control group
 - `anykidsu6` = 1 if any children under age 6
 - `nevermarried` = 1 if never married (all cases are single parents)

1. **(First stage)** Estimate first stage models for the probability of working full time in months 15, 20, 24, and 48, using treatment as the instrumental variable:

$$FT_i(t) = \pi_0 + \pi_1 \text{treatment}_i + \varepsilon_i, \quad \text{for } t = 15, 20, 24, 48$$

What does π_0 represent? What does π_1 represent?

Solution (Part 1). The estimated first-stage regression for each time horizon takes the form:

$$FT_i(t) = \pi_0 + \pi_1 \text{treatment}_i + \varepsilon_i, \quad \text{for } t = 15, 20, 24, 48.$$

Interpretation:

- π_0 represents the *average probability of working full time* among individuals in the control group ($D_i = 0$):

$$\pi_0 = E[FT_i(t) \mid D_i = 0].$$

- π_1 captures the *causal effect of treatment assignment on full-time employment*, i.e. the difference in the average probability of working full time between treated and control groups:

$$\pi_1 = E[FT_i(t) \mid D_i = 1] - E[FT_i(t) \mid D_i = 0].$$

Empirical Results: The table below summarizes the OLS estimates of π_0 and π_1 at each time period t :

Month (t)	π_0	π_1	SE(π_1)	$t(\pi_1)$	N
15	0.1472	0.1374	0.011	12.55	5,480
20	0.1609	0.1147	0.011	10.16	5,245
24	0.1602	0.1074	0.011	9.55	5,224
48	0.2337	0.0506	0.013	4.01	4,796

Discussion: Across all months, the coefficient π_1 is positive and statistically significant at the 1% level, indicating that assignment to the SSP treatment group increased the probability of full-time employment relative to the control group.

The effect is largest at $t = 15$ (about 13.7 percentage points) and declines steadily over time, reaching about 5 percentage points by month 48. This pattern suggests that the SSP program had its strongest impact on promoting full-time work shortly after random assignment, with the effect tapering off in the longer term.

2. **(Reduced form)** Estimate reduced form models for the probability of being on welfare in months 15, 20, 24, and 48, using treatment as the instrumental variable:

$$\text{Welfare}_i(t) = \delta_0 + \delta_1 \text{treatment}_i + \nu_i, \quad \text{for } t = 15, 20, 24, 48$$

where $\text{Welfare}_i(t) = 1$ if person i is on welfare t months since random assignment.

Solution (Part 2). For each horizon $t \in \{15, 20, 24, 48\}$ we estimate

$$\text{Welfare}_i(t) = \delta_0 + \delta_1 \text{treatment}_i + \nu_i,$$

where $\text{Welfare}_i(t) = 1$ if individual i is on welfare t months after random assignment. Here, $\delta_0 = E[\text{Welfare}_i(t) \mid Z_i = 0]$ is the control mean (baseline probability of welfare receipt), and $\delta_1 = E[\text{Welfare}_i(t) \mid Z_i = 1] - E[\text{Welfare}_i(t) \mid Z_i = 0]$ is the *intent-to-treat (ITT)* effect of being offered SSP on welfare status.

t	δ_0	δ_1	$\text{SE}(\delta_1)$	$t(\delta_1)$	$F(\delta_1)$	N	$\Pr[W=1 \mid Z=0]$	$\Pr[W=1 \mid Z=1]$
15	0.810	-0.141	0.0117	-12.06	145.33	5,480	0.810	0.669
20	0.769	-0.121	0.0124	-9.73	94.73	5,245	0.769	0.648
24	0.740	-0.108	0.0127	-8.49	72.09	5,224	0.740	0.632
48	0.592	-0.042	0.0143	-2.94	8.63	4,796	0.592	0.550

Notes. The last two columns equal the model-implied probabilities: $\Pr[W=1 \mid Z=0] = \delta_0$ and $\Pr[W=1 \mid Z=1] = \delta_0 + \delta_1$.

Interpretation. Assignment to SSP substantially *reduced* welfare receipt at all horizons: $\delta_1 < 0$ and statistically significant (1% level or better). At $t = 15$, the control mean is about 81% while the treated mean is about 67%, a decline of ≈ 14 percentage points ($t \approx -12.1$). The ITT effect attenuates over time to ≈ -12 p.p. at 20 months, ≈ -11 p.p. at 24 months, and ≈ -4 p.p. by 48 months (still significant, $t \approx -2.94$). This pattern is consistent with the program inducing earlier exits from welfare that partly dissipate once the subsidy period ends.

3. **(2SLS)** Estimate 2SLS models of the equation:

$$\text{Welfare}_i(t) = \beta_0 + \beta_1 FT_i(t) + u_i, \quad \text{for } t = 15, 20, 24, 48$$

using treatment as an instrument for $FT_i(t)$. Verify that the 2SLS estimate in each case is the ratio of the reduced form and first stage coefficients. *Hint:* use `linearmodels.IV2SLS`.

Solution (Part 3). We estimate, for each horizon $t \in \{15, 20, 24, 48\}$,

$$\text{Welfare}_i(t) = \beta_0 + \beta_1 FT_i(t) + u_i,$$

using treatment_i as an instrument for the endogenous $FT_i(t)$. Identification relies on: (i) random assignment (independence); (ii) instrument relevance (first-stage $\pi_1 \neq 0$);

(iii) exclusion (assignment affects welfare only through FT); and (iv) monotonicity (no defiers). With a binary instrument and treatment, β_1 is the LATE/CACE: the average causal effect of full-time work on welfare among compliers at horizon t .

2SLS estimates. (HC1 standard errors)

Month (t)	β_0	β_1	SE(β_1)	$t(\beta_1)$	N
15	0.960872	-1.026628	0.082099	-12.504746	5,480
20	0.939023	-1.055033	0.105477	-10.002532	5,245
24	0.901966	-1.007911	0.116582	-8.645537	5,224
48	0.785521	-0.829871	0.259718	-3.195279	4,796

Verification. For each t , we verified in the notebook that

$$\hat{\beta}_1 = \frac{\hat{\delta}_1}{\hat{\pi}_1},$$

i.e., the 2SLS slope equals the ratio of the reduced-form coefficient on treatment in the welfare equation ($\hat{\delta}_1$) to the first-stage coefficient of treatment in the FT equation ($\hat{\pi}_1$); the numerical equality holds up to rounding.

Interpretation. The estimates imply a large negative LATE of full-time work on welfare receipt for compliers: at $t = 15$ –24 months, $\hat{\beta}_1 \approx -1$ and highly significant, suggesting that moving from not working FT to working FT (for those whose FT status is shifted by the SSP offer) nearly eliminates welfare participation at those horizons. By $t = 48$, the effect attenuates to about -0.83 and remains statistically significant at conventional levels. This time pattern mirrors the program’s temporary subsidy window and the weakening first stage at $t = 48$. (Note: with a linear probability model, sampling noise can push $\hat{\beta}_1$ slightly outside $[-1, 1]$; the substantive conclusion is unchanged.)

4. **(Characteristics of Compliers)** Find the mean characteristics of the compliers in month $t = 15$: i.e., means of the variables `imm`, `hsgrad`, `agelt25`, `age35p`, `working_at_baseline`, `anykidsu6`, and `nevermarried`. Compare these to the characteristics of always takers and never takers in month 15. *Hint:* define outcome

$$Y_i = X_i \times FT_i(15).$$

Solution (Part 4). For $t = 15$, we compute the mean characteristics of the *compliers* and compare them to those of *always takers* and *never takers*. Following the IV–LATE framework, let Z_i denote assignment to treatment and $FT_i(15)$ the indicator for working full time at 15 months. The groups are defined as:

Always takers: $E[X_i \mid FT_i(15) = 1, Z_i = 0]$, Never takers: $E[X_i \mid FT_i(15) = 0, Z_i = 1]$,

$$\text{Compliers: } E[X_i \mid \text{complier}] = \frac{E[X_i \mid Z_i = 1] - E[X_i \mid Z_i = 0]}{P(FT_i(15) = 1 \mid Z_i = 1) - P(FT_i(15) = 1 \mid Z_i = 0)}.$$

Using the sample shares

$$P(FT_i(15) = 1 \mid Z_i = 0) = 0.148, \quad P(FT_i(15) = 1 \mid Z_i = 1) = 0.285,$$

we obtain the following estimates:

Characteristic	Compliers	Always Takers	Never Takers
imm	−0.026	0.078	0.139
hsgrad	0.117	0.615	0.414
agelt25	−0.058	0.172	0.170
age35p	0.033	0.298	0.327
working_at_baseline	−0.048	0.468	0.122
anykidsu6	−0.145	0.502	0.520
nevermarried	−0.011	0.485	0.483

Interpretation.

- **Compliers** lie between always and never takers, consistent with the IV–LATE interpretation that the SSP affects a marginal subgroup. They are less likely to be immigrants or to have young children, and have lower baseline employment compared to always takers. Their education (**hsgrad**) is lower than that of always takers but higher than never takers.
- **Always takers** are the most advantaged group—older (**age35p**), better educated, and more likely to be working at baseline—implying that they would work full time regardless of treatment.
- **Never takers** are the most constrained—less educated, lower baseline work rates, and more likely to have young children—explaining why they do not respond to the SSP offer.

Overall, the SSP subsidy appears to induce full-time employment primarily among *moderately disadvantaged individuals*—those not as attached to the labor market as always takers, but facing fewer barriers than never takers. This pattern aligns with the local average treatment effect (LATE) interpretation, identifying the program’s causal impact among compliers.